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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech(VLSI Design & Embedded Systems) (CBCS) Degree Examinations

May - June 2024

22VDE233 – SCRIPTING LANGUAGES FOR VLSI

Duration: 3 Hours

Max. Marks:100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	With an example explain push, pop, shift and unshift functions.	10	L*2	1	3
b)	Write a program that computes the circumference of a circle with a radius of 12.5. The circumference is 2 [pi] times the radius, or about 2 times 3.141592654 times the radius.	5	L3	2	3
c)	Modify the program from the previous exercise to prompt for and accept a radius from the person running the program.	5	L3	2	3
2. a)	With an example compare one dimensional array and associative array	10	L2	2	3
b)	Write a program that reads in a list of strings on separate lines and then prints out the list of strings in reverse order - without using reverse on the list.	5	L3	2	3
c)	Write a program that prints a table of numbers and their squares from zero to 32. Try to come up with a way where you don't need to have all the numbers from 0 to 32 in a list, and then try one where you do. (For nice looking output, printf "%5g %8g\n", \$a, \$b prints \$a as a five-column number and \$b as an eight-column number).	5	L3	2	3
3. a)	Demonstrate with an example, VHDL Testbench creation using Perl.	10	L3	1	3
b)	Explain the Working of Netlist Patching.	5	L2	1	3
c)	Write a procedure for test vectors generation in Perl.	5	L2	1	3
Unit – II					
4. a)	Explain Tcl structures and syntax? List the summary of Tcl.	10	L2	3	3
b)	Explain variable substitution with an examples.	5	L2	3	3
c)	Explain the switch command.	5	L2	3	3
5. a)	Write a procedure syntax in Tcl, and also write a program to demonstrate procedure command.	10	L2	3	3
b)	Explain the while command.	5	L2	3	3
c)	Write a Tcl program to read data from the channel.	5	L3	3	3
6. a)	Explain Eval command in Tcl with an example.	10	L2	4	3
b)	Make use of globe command to return list of files that match the *.tcl pattern.	5	L2	4	3
c)	Make use of Tcl/Tk to develop graphical program to display Hello World!	5	L3	4	3
Unit – III					
7. a)	With a neat diagram, explain the specification driven design decomposition.	10	L2	5	3
b)	Explain Register Transfer Level. State Disciplined User Principle.	5	L2	5	3
c)	State Fundamental Verification Principle and Retain Useful Information Principle.	5	L2	5	3
8. a)	With a neat diagram, explain hybrid random simulation environment.	10	L2	5	3
b)	Explain the synthesizable testbench.	5	L2	5	3
c)	State and describe Orthogonal Verification Principle.	5	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
